NUS National University of Singapore

BT5110
Data Management and Warehousing

Introduction

Class Information

- **Lecturer:** Lee Mong Li (dcsleeml@nus.edu.sg)
- **Tutors:** Chen Kaiwen (kaiwenchen66@u.nus.edu)
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- **Tutorials start on Week 3:**
  G1, G2, G3 on Wednesdays (20:30 - 21:30)
  G4, G5, G6 on Fridays (16:00 – 17:00)
- **Course website: CANVAS**

Assessment – 100% CA

- **Written Assessment**
  - Week 7 – Quiz (20%)
  - Week 11 – Test (30%)
- **Practical Assessment**
  - Week 5 – Conceptual design (10%)
  - Week 8 – Schema refinement (10%)
  - Week 13 – Dashboard demo and final report (25%)
- **Tutorial Attendance and Participation** (5%)

Academic Integrity

- **Zero-tolerance**
  - Cheating, fabrication, plagiarism and violation of intellectual property or copyright are taken seriously
  - Academic misconduct may result in disciplinary action
- **Unintentional plagiarism**
  - Protect your project work and do not make it unnecessarily accessible to others
  - You may also be held responsible if your work enables another student to plagiarise
- **Self-plagiarism**
  - Do not reuse work submitted for another module and present it as new work for this module

Online Resources

- The Internet provides access to a vast range of resources
- This is especially common in programming, where code is frequently shared online
- **Being publicly available does not mean material is free to copy or reuse without acknowledgement**
- **Any code, text, ideas, or other material you use must be properly acknowledged**

AI Tools

- The same academic integrity principles apply to AI tools
- **Acknowledge your use of AI**
- **You remain responsible for your work**
  - Understand and verify anything generated by AI
  - Do not submit AI-generated content that you cannot explain or defend
- **AI output may be unreliable**
  - Responses depend on the prompt and context provided
  - Information may be incomplete, outdated or incorrect
  - References, citations, data or code may be fabricated or inaccurate

NUS Guidelines on the Use of AI Tools
https://libguides.nus.edu.sg/new2nus/acadintegrity#s-lib-ctab-22144949-5

Do Not Redistribute NUS Course Materials

- Course materials on Canvas or any approved platforms are provided **for course participants only**
- **Do not share, upload, repost, or redistribute** course materials to third parties
- This includes slides, notes, recordings, assessments, solutions, and other teaching materials
- Redistribution requires **explicit permission from the module instructor or authorised NUS personnel**

What is Data Management?

- The **systematic process of collecting, storing, organizing and maintaining data** to ensure data quality, availability and security throughout its life cycle
- Key elements of data management:
  - Data governance – Setting policies and standards
  - Data integration – Combining data from diverse sources
  - Data quality – Ensuring accuracy, completeness and consistency

Data management serves as the foundation for deriving value from data, enables analytics, machine learning, and informed business strategies.

What is a DBMS?

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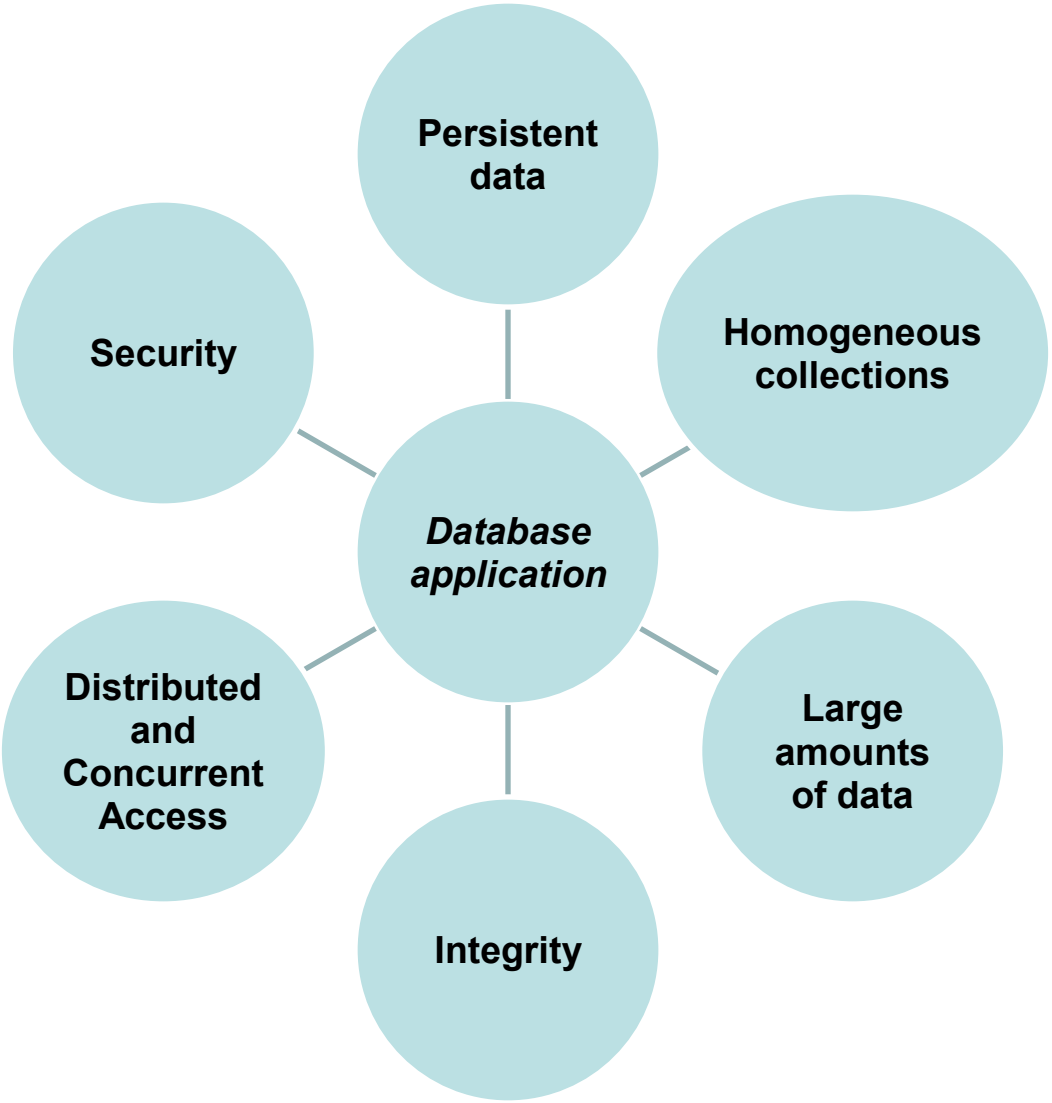
- Database Management Systems (DBMS) is a **core technology for implementing data management**
- Key aspects of DBMS
  - **Data storage** – Provide structures for organizing and storing data
  - **Data access and retrieval** – Enable efficient querying and manipulation of data
  - **Data quality enforcement** – Use integrity constraints to maintain data accuracy and consistency
  - **Security and privacy** – Control who can access and modify data through user roles and privileges
  - **Backup and recovery** – Protect against data loss and restore the database after failures

DBMS Applications

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- A DBMS is a generic platform for the development and management of database applications
  - Examples: MySQL, PostgreSQL, Oracle
- A database application is a **collection of data** together with **the programs that allow users to access and manipulate the data**
  - Banking – accounts, transactions
  - University – student registrations, grades
  - Airline – reservations, schedules
  - Sales – customers, products, purchases
  - Manufacturing – inventory, orders

DBMS Features



Persistent Data

- Data must persist
- How can data outlive the process that created it, and be reused by other processes?
  - Enable data continuity and availability
- Volatile vs. persistent memory: trade-off between speed, cost and capacity

Memory Type	Volatility	Example	Use Case
Primary Memory	Volatile	RAM	Fast access during active processing
Secondary Memory	Persistent	SSD, HDD	Long-term storage, files, databases
Tertiary Memory	Persistent	Magnetic tape, Optical media	Backup, archival, large-scale data storage

Large Amounts of Data

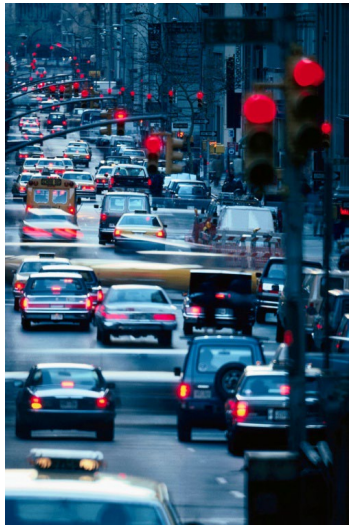
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- Databases are designed to handle **data that may not fit entirely in memory**
- External storage provides larger capacity, persistence, and lower cost per unit of storage
- When data resides on external storage, **I/O often becomes the dominant cost**
- How do we process data that does not fit in memory?
  - Example: India has over 1.1 billion mobile phone subscribers, second-largest user base in the world
  - For each subscriber, telecom companies store data such as *Subscriber Name, Mobile Number, Region/State, Plan Type and Usage History*

Homogeneous Collections

- Database applications manage **collections of similar entities**: students, customers, products, flights
- Data is represented in a consistent structure → enables efficient querying and indexing





Integrity

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- How do we maintain the integrity of data despite possible application, system or media failures?
- Restore data to a consistent state after failures
- A **transaction is a logical unit of work**
  - Example: Transfer $100 from account A to account B
  - A transaction consists of debiting A and crediting B
- **Recovery ensures database consistency after failures**
  - **Atomicity**: All actions in a transaction happen, or none happen
  - **Durability**: Effects of a successful transaction persist

Concurrent Access

- Modern databases support millions of users
  - Example: Global Distribution Systems like Amadeus and Galileo handle flight bookings, hotel reservations and car rentals with many users at the same time
- **Concurrency control ensures consistent data access and updates**
  - **Isolation:** Concurrent transactions should not interfere with each other
  - **Consistency:** Transactions must preserve the validity of the database, e.g., two users should not both be able to book the last available seat

ACID = Atomicity • Consistency • Isolation • Durability

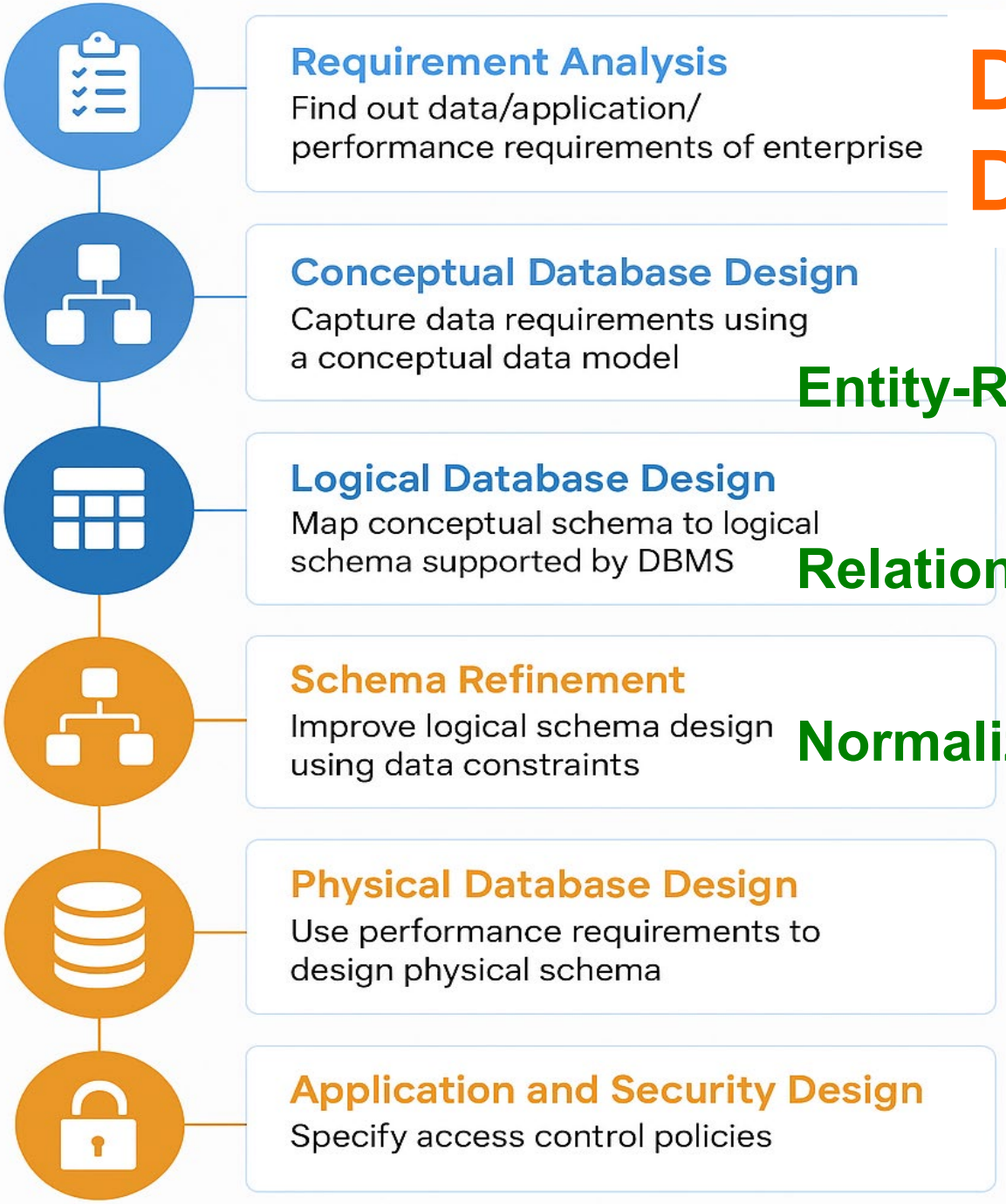
Security

- Most real-world database applications handle sensitive, mission-critical data
  - Example: education, healthcare, finance, logistics
- Important to **protect data from unauthorized access and modification**
- DBMS supports **fine-grained control** over
  - Who can read the data?
  - Who can modify the data?
  - Who can grant access to others?

Data Models

- **A data model is a set of constructs to organize and describe data**
  - Example: Relational model (tables), Document model (JSON-like documents), Graph model (nodes and edges)
- **Data Manipulation Language (DML)**
  - Operations to retrieve, insert, update and delete data
  - Different database models provide different query languages or APIs
  - Example: Relational → SQL, Document → MongoDB query operations

Database Design Process



Entity-Relationship (ER) model

Relational model and SQL

Normalization

Data Warehousing

- **A data warehouse integrates data from multiple sources into a centralized repository for analytics and decision-making**
- Key characteristics
  - **Centralized repository** – Aggregates data from multiple sources
  - **Subject-oriented** – Organized around key business themes such as customers, sales, and inventory
  - **Time-variant** – Stores historical data for trend analysis
  - **Non-volatile** – Data is mainly read for analysis; updates typically occur through ETL
- **ETL (Extract, Transform, Load)** move and prepare data from operational systems for the data warehouse

Database vs Data Warehouse

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	Database	Data Warehouse
Purpose	Support day-to-day operations (OLTP)	Support analytics and decision-making (OLAP)
Data	Current operational data	Integrated historical data
Workload	Frequent, short transactions	Complex analytical queries over large volumes of data

Database → Run the business
Data Warehouse → Analyze the business

Course Focus

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- **Database Design**
  - How to model the data requirements of applications?
  - How to organize data in databases?
  - Topics: Relational Model, ER Model, Schema Refinement
- **Database Programming**
  - How to create, query and update databases?
  - How to specify and enforce data constraints?
  - Topics: SQL, Triggers & Stored Functions, NoSQL
- **Data Warehousing**
  - How to organize data for analytics and decision-making?
  - Topics: Dimensional Modeling

Course Schedule

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Week	Date	Lecture	Tutorial
1	12-Aug	Introduction	
2	19-Aug	Relational Model	
3	26-Aug	Conceptual Modeling (ER)	T1 – Relational Model (PostGreSQL)
4	2-Sep	Schema Refinement I	T2 – ER model
5	9-Sep	Schema Refinement II	T3 – FDs
6	16-Sep	SQL I	T4 – Normalization
	23-Sep	RECESS	
7	30-Sep	SQL II / Quiz	T5 – Project checkpoint
8	7-Oct	SQL III	T6 – SQL (Simple)
9	14-Oct	Triggers and Stored Functions	T7 – SQL (Aggregate + Nested)
10	21-Oct	Data Warehousing	T8 – SQL (Nested + Triggers)
11	28-Oct	Test	T9 – Project checkpoint
12	4-Nov	NoSQL	T10 – JSON and MongoDB
13	11-Nov	Project Presentations	Dashboard Demo, Final Report

Databases in the Age of AI

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AI can help with…	You still need to understand…
Writing SQL queries	How to design databases that meet business requirements
Generating database schemas	How to model data, relationships, and constraints correctly
Producing database code	How to refine schemas to eliminate redundancy and maintain data integrity
Summarizing data	How data warehouses support analytics and decision-making

Learn databases not because AI cannot write SQL, but because you need to understand whether the data design and results are correct.